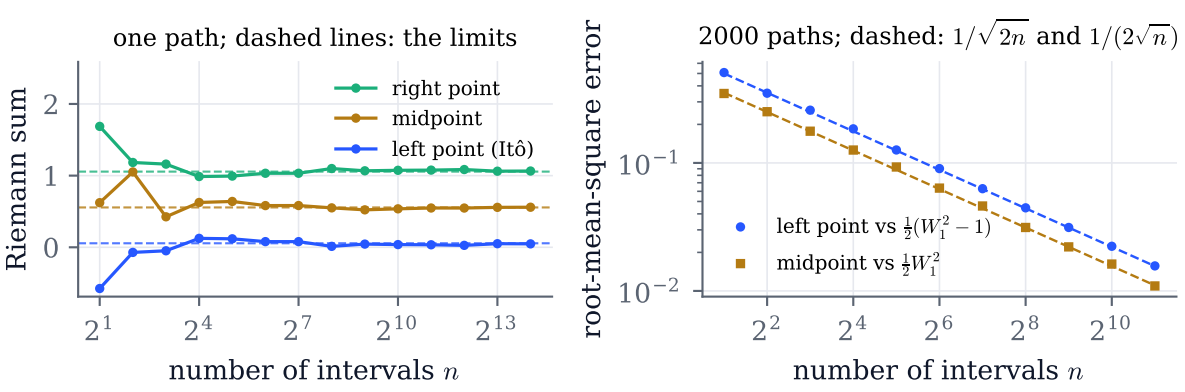
one path; dashed lines: the limits
Riemann sum
2
1
0
right point
midpoint
left point (Itô)
2^1
2^4
2^7
2^{10}
2^{13}
number of intervals n
2000 paths; dashed: $1/\sqrt{2n}$ and $1/(2\sqrt{n})$
root-mean-square error
10^{-1}
10^{-2}
left point vs $\frac{1}{2}(W_1^2 - 1)$
midpoint vs $\frac{1}{2}W_1^2$
2^2
2^4
2^6
2^8
2^{10}
number of intervals n